Software Development Project — G&G City · Lab Backlog Sprint 1

LAB S1: Station Backgrounds — Mohammed Al-Mandili

Topic: Designing Consistent Scene Backgrounds for a Sliding Web Game

Learning outcomes

By the end of this lab, students will be able to:

- Design four backgrounds that look like one city
- Export images at a fixed size and a web-friendly weight
- Leave a clear stage area for characters

🎯 Problem Scenario

The stations slide next to each other like pages in a book. If one background is a different size or its floor is higher, the jump is visible and the cat floats.

- **S2 (Aws) shows your images with `background-size: cover` in a 16:9 screen**
- **S3 (Ayman) places the cat and the box on the floor of each station**

Your files: `scenes/workshop.png`, `scenes/backyard.png`, `scenes/charging.png`, `scenes/repo.png`

Required Outputs

Build:

1. **The workshop first (`workshop.png`): S2 and S3 test with it**
2. **Backyard, charging station, and the GitHub main repository (`backyard.png`, `charging.png`, `repo.png`)**
3. **Every image: `1920 × 1080` PNG, under 1 MB**
4. **The bottom third of every image is empty floor at the same height**

Code is given only for the hard parts. Parts marked 📝 **Note** are easy: follow the note and write them yourself. Test with **Live Server** in VS Code (`fetch` does not work when you double-click a file).

Model Answer:

Task 1: The workshop

> 📝 **Note (no code needed):** Draw a workbench, tools on the wall, and an open space in the middle. Keep important details in the top two thirds. Push it early, even as a draft: everyone else is waiting for it to test.

Task 2: The other three stations

> 📝 **Note (no code needed): Backyard** (where files wait), **Charging station** (where the cat recharges between commands), **Main repository** (a big building that represents GitHub — your own design, no GitHub logos). Use the same color palette, light direction and camera angle in all four.

Task 3: Export settings (the part people get wrong)

> 📝 **Note (no code needed):** Canvas size exactly `1920 × 1080` before you start drawing, not resized after. Export as PNG. If a file is over 1 MB, compress it at **tinypng.com** or export at 8-bit. File names are lower-case with no spaces, exactly as listed.

Task 4: Check the floor line

> 📝 **Note (no code needed):** Open the four images side by side and draw a horizontal guide at about 65% of the height. The floor edge must sit on that guide in every image. Then open `index.html` with Live Server and check the cat stands on the floor.

Submission

```
git pull
git add scenes/
git commit -m "إضافة خلفية الورشة"
git push
```